How System Information Is Found

What 44helper reads about the Pi and its software, how each reading is made, which pieces of software the tabs look for, and what has to be installed before each part of the setup works.

PLATFORM PRODUCTS READINESS BOOT REPORT

SUMMARY · READ-ONLY REVIEW

Where the readings come from

The helper gathers its facts in four groups: the platform (the board, the OS and the network), the installed products, the tunnel and firewall state, and the readiness of each setup step. The Boot report reads a fifth group, the boot record.

GROUP	READ BY	SHOWN ON
Platform	`system_profile`, cached once per run	Header text, default distro, Ampersand check, Boot report
Installed products	dpkg, the unit files and the program files	Overview, Services, the Config tab's status
Tunnel and firewall	wg0, the wg0 config, the 44NetConnect zone	Overview, Full and Split tabs, the self-check
Step readiness	One status function per setup step	Overview and each tunnel tab
Boot record	journalctl, timedatectl, vcgencmd, the boot log	Boot tab and `--boot-report`

The Overview is served from one route, which reads all four groups together. Passing `force=1` refreshes the installed-product cache before it reads.

The board, the OS and the network

2.1 How it is read

The platform reader is the shared function `system_profile`, copied unchanged from `shared/confio.py` (see the Common Functions Sheet). It is called once, the first time the helper needs it, and the result is kept for the life of the process.

FIELD	WHERE IT COMES FROM
board, model	`/proc/device-tree/model`. A board is "pi" if the model names a Raspberry Pi, otherwise "other".
os_id, os_name, codename, version_id	`/etc/os-release`: ID, PRETTY_NAME, VERSION_CODENAME and VERSION_ID.
arch	`dpkg --print-architecture`, or the machine name from `uname` if dpkg is not there.
wifi_manager	NetworkManager if its service is active; else wpa_supplicant if its socket folder has files; else systemd-networkd if active; else "unknown".
wifi_iface, wifi_ifaces	The wireless interfaces under `/sys/class/net`. The one that is up is preferred; else wlan0; else the first one.

If the reader fails, the helper uses a blank platform with the board set to "other". The helper still runs.

2.2 What the platform is used for

- **The default distro** for install steps is the OS codename if it is bookworm or trixie; otherwise trixie. The install tabs can still be set to the other release.
- **The platform text** in the header reads the OS name, the architecture and the board.
- **Ampersand** checks, on the node itself with a shell line, that VERSION_CODENAME is trixie before anything else is done (Stage 1, Part 5.5).

The cache is kept for the life of the process. After an OS upgrade the helper must be restarted before it reports the new codename.

What is on the node

3.1 The catalogue

The helper keeps a list of the products it knows, and for each one, the packages, the program files, the systemd units and the leftover paths that show it is there:

PRODUCT	PACKAGES (MATCHED BY NAME)	PROGRAM FILES	UNITS
ASL3	asl3, asl3-asterisk, allmon3	none listed	asterisk, allmon3
DVSwitch	dvswitch-server, analog-bridge, mmdvm-bridge, md380-emu	/opt/MMDVM_Bridge/MMDVM_Bridge, /opt/Analog_Bridge/Analog_Bridge	analog_bridge, mmdvm_bridge, md380-emu
USRP2M17	none (matched by name pattern)	/opt/USRP2M17	usrp2m17
SVXLink	svxlink-server (and names starting svxlink, remotetrx or qtel)	/usr/bin/svxlink, /usr/local/bin/svxlink	svxlink, svxlink-usrp, remotetrx
Ampersand	names starting amp-	/usr/bin/amp-server, /usr/etc/amp-hub.env	amp-server, amp-hub

3.2 How it is read

1. One `dpkg-query` call lists every package with its status.
2. For each product, its packages are taken from that list. A package counts when its status says it is installed.
3. Its program files are checked with a file-exists test.
4. Its units are read with one `systemctl show` call, giving the load state, the active state and the unit-file state. A unit that is masked is flagged.
5. Leftover paths and removed packages with kept settings are noted.

The whole result is kept for 20 seconds. A refresh with `force=1` bypasses it.

3.3 The states

STATE	MEANING
installed	A core package, a core program file, or a core unit is present.
broken	A package is half-installed (its status is not a clean install), or the package is installed but its working parts are missing (for example the package is there but the expected file is not). The note sends the operator to the Update tab's package recovery steps.
leftovers	Nothing is installed, but a package has its settings kept, or a leftover path, a program file or a unit file remains.
absent	None of the above.

A product found only by its program file is noted as "found by file", not as a package install. A masked unit is listed in the notes.

What the helper checks about the tunnel

- **wg0 present:** the interface's folder exists under `/sys/class/net`.
- **wg0.conf permissions:** the file must be mode 600. If not, the self-check warns that the private key's permissions have drifted.
- **The 44NetConnect zone:** its ports and services, read from firewalld (cached). If the zone does not exist yet, the self-check says that is expected before the tunnel's steps 2 to 6.
- **Exposure:** the self-check looks for the helper's own port, the other dashboard ports it watches, and the Asterisk AMI port in that zone. Any of them exposed is a danger finding. A clean zone gives an "ok" finding.
- **Configured tunnels:** the names of the tunnels in the config (for example the Pi's tunnel and any router tunnel).

The self-check looks at exposure, not at what is installed. Installation is read in Part 3 and Part 5.

Whether each setup step is done

Each step of the tunnel setup has one status function that answers "done" or "not done", with a few facts behind the answer. The Overview combines them. The step numbers match Stage 2.

STEP	DONE WHEN
1. Firewall prerequisite	firewall-cmd is installed and firewalld is active
1a. systemd-resolved	systemd-resolved is active and answers DNS queries (the answer is cached)
2. The 44NetConnect zone	The zone exists
3. Inbound services and ports	The IAX2 port is open in the zone. The IAX2 port comes from the config (4569/udp by default). The check also accepts the port being listed, not only the service name.
4. Config pasted	wg0.conf exists and passes the checks in Stage 2, Part 2
5. wg0 attached to the zone	The zone lists wg0
6. Tunnel enabled and started	wg reports a latest handshake for wg0. The check is the handshake itself, not only that the unit is running.
DNS, AllowedIPs, reflector egress, boot order	Each has its own status: the DNS settings, the AllowedIPs shape for the tab, the egress ports, and the boot units

A step can be done in the record but broken on the node (for example a zone that exists but no longer lists wg0). The status answers from the live system where it can, not from the step's own record.

The requirements, by setup

NEEDED FOR	PACKAGE OR PROGRAM	CHECKED BY
Any install step	A healthy package system (dpkg reports no problems)	The package-health gate (Stage 1, Part 6)
The tunnel's firewall	firewalld	Step 1
The tunnel's DNS	systemd-resolved, wireguard-tools	Step 1a
A DNS = line in the config	resolvconf (from openresolv, or the systemd-resolved shim)	The DNS step refuses without it
A full tunnel	nftables, or iptables-restore, for wg-quick's routing rules	Step 6 refuses without one
Starting the tunnel	The wg-quick@ template. Without it, the helper's fallback unit runs wg-quick up directly.	Step 6 chooses the unit
DVSwitch	The DVSwitch packages (Stage 1, Part 5.2)	The DVSwitch product state
SVXLink	svxlink-server or the USRP build (Stage 1, Part 5.3)	The SVXLink product state
M17	git, PHP for Apache, and USRP2M17 from its installer	The USRP2M17 product state
Ampersand	Debian 13 (Trixie), and the amp packages	The shell check, and the Ampersand product state
Remote control	Node.js 22 and Claude Code	The Claude Remote steps
Cloudflare connector	cloudflared	The Cloudflare steps

The requirements for each tab's installs are in Stage 1. The list above is what the helper checks before it lets a step run, not a full install list.

The boot record

7.1 What it reads

PART	SOURCE
Host, node, boot time and uptime	The boot ID, `/proc/uptime`, the host name
Clock	`timedatectl`: whether NTP is on and synchronised, and the time zone
Power	`vcgencmd get_throttled`
Journal	`journalctl --list-boots`, and this boot's warnings
Boot order and the boot log	The boot-order units and `/var/lib/44helper/boot.log`
The last report, the tunnel and its guard	The saved reports, the tunnel config, and wifimon's tunnel guard state
Versions	Each tool's version

The report is built as text, or as JSON with `--json` (Stage 1.5). Its findings are warnings about things it can see. The manual lists the kinds, such as a restarted unit, a service that never started, an unsynchronised clock, or under-voltage.

7.2 Snapshots

A background loop saves a snapshot of the boot record 6 minutes after each boot and updates it at 25 minutes, one file per boot. Each snapshot is kept to 64 KB, and the snapshots are listed under "Saved copies of earlier boots" on the Boot tab.

What the readings show

8.1 The readings are read-only

Every reader in this summary only reads. The book can say that looking at the state changes nothing on the node.

8.2 Some readings are cached

The platform is cached for the life of the process, the products for 20 seconds, and the resolver answer for a while. A reading can be a little old, and a restart or a forced refresh brings it up to date.

8.3 Product detection is by name and by file

A product is found by its package name pattern and its program files. A hand-built program with no package is found only if its file is in the list. The book should say which products are matched by package and which by file.

8.4 The self-check checks exposure

It does not check that the tunnel works, nor that the packages are installed. Those are the readiness steps in Part 5.

8.5 Not checked

- The Nodes tab, which reads the ASL node's own state through Asterisk.
- The router probes, which connect over SSH to the router.
- The exact dpkg and systemctl output on a live node. The parsing was read against the expected formats.